

# Unsupervised Learning - Lab 08 - Bayesian Networks

## An Introduction

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# Agenda

1. Load the Asia Bayesian Network from a BIF file.
2. Visualize the network structure using NetworkX and Graphviz.
3. Display Conditional Probability Tables (CPTs) for selected variables.
4. Query the network to find parents, children, and Markov Blanket of specific nodes.
5. Test d-separation for pairs of variables to understand independencies.
6. Generate synthetic data from the Bayesian Network using forward sampling.

# Introduction to Bayesian Networks

- Bayesian Networks are probabilistic graphical models that represent a set of variables and their conditional dependencies via a Directed Acyclic Graph (DAG).
- Today, we will use the `pgmpy` library to work with the Asia Bayesian Network.

## Loading the Asia Bayesian Network

- The Asia network is a small Bayesian network that is often used for educational purposes.
- We will load this network from a BIF file using `pgmpy`'s `BIFReader`.
- `BIFReader` reads the network structure and parameters from the file.

- Visualization helps in understanding the structure of the network.
- We will use `NetworkX` and `Graphviz` to visualize the Asia network.
- `NetworkX` provides a simple visualization, while `Graphviz` offers a more polished layout.

## Conditional Probability Tables (CPTs)

- Each node in a Bayesian network has a CPT that defines the probability distribution of the node given its parents.
- We will display the CPTs for selected variables to understand their dependencies.

## Interacting with the Graph

- We can query the network to find the parents, children, and Markov Blanket of specific nodes.
- This helps in understanding the relationships and dependencies between nodes.

- D-separation is a criterion for deciding whether a set  $X$  of nodes is independent of another set  $Y$  of nodes given a third set  $Z$ .
- We will test d-separation for pairs of variables to understand their independencies.

# Sampling from the Bayesian Network

- Bayesian networks can be used to generate synthetic data.
- We will use forward sampling to generate data from the Asia network.
- This data can be used for further analysis or testing.

- pgmpy documentation: [pgmpy.org](http://pgmpy.org)
- Asia network BIF file: [BN Repository](#)